

GCC Growth and Diversification: The Role of Cross-Border Investment and Sovereign Wealth Funds*

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Abstract

This paper investigates the role of Gulf Cooperation Council (GCC) cross-border investments, including by Sovereign Wealth Funds (SWFs), in driving economic growth and diversification of GCC countries. Following the pandemic, GCC countries have accelerated reforms to attract foreign investment and talent. These efforts align with their broader growth strategies, aiming to enhance productivity, diversify their economies, and support sustainable growth alongside the energy transition. Utilizing a novel deal-level dataset, we analyze GCC cross-border investment portfolios across various dimensions such as time, geography, sector, and industry. We show that recent years have witnessed an increase in GCC cross-border investments. While the geographic distribution of these investments remains diverse and balanced across different regions, both inward and outward investments are increasingly directed towards the services sector. The empirical results demonstrate a positive impact of both inward cross-border investments and domestic investments on GCC real non-hydrocarbon GDP growth. Notably, the impact of inward investments is almost three times larger in the medium term. This amplification could be explained by increased investments in high-growth, higher-value added sectors, knowledge-intensive sectors, and more recently, renewable energy. This paper highlights the ongoing economic transformation in GCC countries, driven by strategic investments. SWFs are emerging as key contributors to this transformation.

Keywords: Foreign Direct Investment, Economic Diversification, Green Economy

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*The views expressed are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

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I Introduction

Gulf Cooperation Council (GCC) countries are actively diversifying their economies across output, export, and revenue dimensions, aligning with their economic strategies and global decarbonization efforts. Nevertheless, GCC countries still face challenges in achieving higher levels of economic diversification and foreign investment, particularly inward foreign direct investments (FDI) directed towards non-hydrocarbon sectors with high growth potential.

The GCC region boasts some of the world's largest sovereign wealth funds (SWFs), with 13 GCC SWFs¹ collectively managing over \$4.2 trillion in assets. These SWFs were established to ensure sustainable long-term economic growth, promote inter-generational equity, and mitigate risks associated with fluctuating hydrocarbon prices. More recently, GCC SWFs have focused on larger, higher-profile, and more active strategic investments, through acquiring, planning to acquire, or investing in a major global institution or firm, or financing large strategic projects domestically. This approach can signal a more active role for SWFs in the broader industrial strategy of the GCC countries, facilitating the development of joint ventures (joint stock companies), fostering the creation of strategic alliances, and easing knowledge sharing, technology transfer, or the establishment of industrial complementarities that benefit the home country.

In this context, SWFs have been making significant investments (both domestic and foreign) and establishing strategic alliances in technology-driven and high-value-added sectors. These partnerships often involve collaboration with Advanced Economies (AEs). For example, Saudi Arabia's Public Investment Fund (PIF) has committed to SoftBank's Vision Fund, which focuses on emerging technologies like artificial intelligence (AI) and robotics. Similarly, the UAE has established the G42 group to drive AI research and deployment, while Mubadala Investment Company is investing in the semiconductor industry. Moreover, in their efforts to facilitate the

¹Namely Mumtalakat Holding (BH), Kuwait Investment Authority (KW), Oman Investment Authority (OM), Qatar Investment Authority (QA), Public Investment Fund (SA), Abu Dhabi Investment Authority (AE), Investment Corporation of Dubai (AE), Mubadala Investment Company (AE), Abu Dhabi Developmental Holding Company (AE), Emirates Investment Authority (AE), Dubai Investment Fund (AE), Sharjah Asset Management (AE), Fujairah Holding (AE), and Mubadala Development Company (AE).

transition to green and sustainable economies, PIF has launched NEOM, a \$500 billion smart city project in Saudi Arabia with a focus on renewable energy, including solar and wind, featuring innovative energy-efficient infrastructure and technologies. Likewise, Mubadala has established Masdar, a renewable energy and sustainable technology company, to promote clean energy projects in the UAE and globally. Notably, Masdar has invested in various renewable energy initiatives in more than 40 countries globally. Furthermore, UAE has also signed a new Partnership for Accelerating Clean Energy (PACE) with the USA which is set to invest \$100 billion in a new clean energy framework.

Against this backdrop, this paper investigates the role of cross-border investments and those by SWFs in promoting economic growth and diversification within GCC countries. We leverage novel deal-level data on mergers and acquisitions (M&A) from the BvD Zephyr database and data on investments by SWFs from the Sovereign Wealth Funds Institute (SWFI) database. Utilizing this data, the paper first provides a comprehensive overview of the investment landscape across GCC economies during 2000-2023 period. This overview encompasses domestic, inward, and outward investments and considers them across multiple dimensions, including time, geographic distribution, sector (public vs. private), and industry (oil vs. non-oil).

More importantly, by analyzing the domestic and cross-border investment portfolios (including SWFs' investment), the paper sheds light on their role in GCC economic diversification efforts. The study investigates whether GCC and GCC SWFs' investment, both domestic and cross-border, have spurred growth in non-hydrocarbon sectors. The international coverage of the data allows for a better understanding of GCC and GCC SWFs' international investment portfolios. This analysis helps identify potential common risk exposures, such as concentration in specific countries or regions and/or by industry, and the strategic nature of these investments. Exploring these different dimensions will help facilitate policy discussions on how to enhance GCC growth models and further support GCC economic diversification and promote green transition.

The paper finds that: (i) recent years have witnessed an increase in both inward and outward cross-border investments by GCC countries; (ii) the geographic distribution of GCC cross-border investments is becoming more diverse and balanced;

(iii) since the global financial crisis, GCC SWFs' share of outward and domestic investments has been gradually increasing; (iv) both inward and outward investments increasingly target the services sector, particularly transportation and logistics, information and communication, real estate activities, professional, scientific and technical activities, and hospitality and tourism; (v) recently, GCC countries have also been increasing their share of investments in clean and renewable energy, with the UAE and Saudi Arabia leading this trend; (vi) inward and domestic investments have a statistically significant positive effect on real non-hydrocarbon GDP growth, with inward investments exhibiting a larger impact. Specifically, a one percentage point increase in the ratios of inward and domestic investment to GDP is associated with an estimated increase of 1.2 percent and 0.4 percent in real GDP after five years, respectively.

II Literature review

This paper relates and contributes to three strands of literature. First, this paper contributes to the literature on the relationship between economic diversification, growth, and FDI. Many studies have found a positive correlation between FDI inflows and economic diversification. For instance, Iwamoto and Nabeshima (2012) found that FDI inflows are positively correlated with export diversification and sophistication, and FDI stocks contribute positively to export sophistication, using a dynamic panel data model. Similarly, Tadesse and Shukralla (2011) found a positive correlation between FDI stocks and horizontal diversification of exports, applying parametric (quantile) and semi-parametric econometric methods on a cross-country sample. As for country- and region-specific studies, Amighini and Sanfilippo (2014) concluded that FDI fostered diversification in low-tech industries and enhanced manufacturing exports' quality in African economies, and Lee, Fariza and Sharipova (2015) focused on the case of Kazakhstan and found a positive relationship between FDI and economic growth.

These findings are supported by theoretical arguments as well. According to Dunning's eclectic paradigm, FDI can stimulate economic diversification by leveraging host country-specific advantages, including human capital, infrastructure, and institutional quality.

While existing literature provides robust empirical evidence and theoretical underpinnings for a positive relationship between FDI and economic diversification, few studies have focused on the role of FDI in economic diversification in oil-rich countries, especially in GCC. For example, Hussein (2009) found a weak relationship between FDI and GDP in the panel of the GCC countries using country-level data, and IMF (2014) found that FDI inflows into the GCC have not been associated with sectoral improvements using sectoral FDI data. To the best of our knowledge, this is the first paper that leverages deal-level and project-level investment data to provide a comprehensive view of the GCC's cross-border investment portfolio, including both inward and outward investment. This allows more granular empirical analysis on the relationship between cross-border investments and economic growth and diversification at the industry level.²

Second, this paper contributes to the literature on the relationship between economic diversification and SWFs. Mami (2023) conducted a systematic meta-narrative review of the role of SWFs in natural resource-rich countries and concluded that they have an important role in supporting industrial development and economic growth. However, to the best of our knowledge, there have been no studies that quantitatively examine the role of SWFs in facilitating economic diversification and driving economic growth in the resource-rich countries like the GCC. Therefore, this paper contributes to this strand of literature by providing quantitative empirical evidence on the role of SWFs in the GCC's economic diversification, using granular deal-level and project-level SWFs investment data.

Third, this paper contributes to the literature on the impact of knowledge-intensive, high-value added, and green investments on economic growth in GCC countries. IMF research (2023b) highlights a significant multiplier effect, with ICT investments showing a 1.8x multiplier on non-oil GDP growth (roughly double the impact of non-ICT investments). This underscores the potential for ICT investments to significantly support economic diversification in the GCC. In contrast to the oil sector, where investment multipliers are negligible, recent analysis suggests that green and sustainable infrastructure investments have output multipliers ranging from 1.1 to 1.5, ex-

²The majority of cross-border deals in the database likely meet the 10% ownership threshold required to qualify as foreign direct investment (FDI) according to the Balance of Payments Guide. However, we refrain from using the FDI definition formally in this analysis because we haven't conducted a comprehensive FDI classification analysis.

ceeding traditional infrastructure investments with multipliers between 0.5 and 0.6. Additionally, green energy investments appear to be more labor-intensive compared to brown energy investments. This suggests the potential for green investments to create more jobs. Garrett-Peltier (2017) found that shifting \$1 million from brown to green energy creates a net increase of 5 jobs. Similar conclusions are drawn by other studies (Hepburn et al., 2020; Engel & Kammen, 2009). The IMF (2023b) further suggests that combining green investments with the full implementation of reforms under the UAE’s 2050 strategies could nearly double the UAE’s potential long-term non-oil GDP growth. Similar results and magnitudes were found for Saudi Arabia (IMF, 2022).

The rest of the paper is organized as follows. Section III describes the sample and data sources. Section IV presents the empirical methodology deployed to estimate the growth effect of investment (including SWFs’ investment). Section V discusses the results, including stylized facts about GCC inward and outward investment as well as their SWF-involved investment, and empirical results of their growth effects. Section VI concludes.

III Data

The main data sources for this paper are the BvD Zephyr and SWFI databases. The BvD Zephyr database consists of deal-level data on M&A, joint ventures, investment holdings, and financing transactions for most countries in the world during the two-decade period from 1990s until now.³ They contain detailed information on the deals, such as, the information on the parties involved, including targets, acquirers and vendors, amounts, types of deals, status, etc. For example, the databases include information on the types, shareholder structures (including public versus private), and industries of the targets, acquirers and vendors, which enables us to construct a comprehensive view of the GCC’s country-wide and SWFs’ investment portfolios and conduct granular empirical analysis with sufficient variation at the dis-aggregated level.

In the data cleaning process of the BvD Zephyr data, we retain only deals with a

³SWFI’s data goes back to 1970s.

status of either completed, announced, or pending. For the announced and pending statuses, we further require that the last data update is no more than 10 years old. Therefore, deals are dropped if they have been pending or announced (not yet completed) for more than ten years. As a rule, deal values are available for most deals. However, for deals lacking a value, we estimate the deal value using information on the share acquired and the total equity of the company being acquired. Overall, after cleaning, our sample consists of 10,058 transactions (domestic and cross-border) for the GCC countries, including 1419 deals financed by SWFs, for the 2000-2023.⁴

SWFI platform offers data on a variety of institutional investors including SWFs, pension funds, asset owners, and more. We limited our selection to deals that directly mention GCC SWFs as acquirers or have government specified as a buyer type. In total, after cleaning, the sample of deals consists of 3,656 unique transactions for GCC countries for the 2000-2023 period.⁵

Next, within the universe of the deals, there are some deals that are wholly domestic, for example, a Saudi Arabian company acquiring or investing in another Saudi Arabian company. Some are wholly cross-border, such as a German company acquiring or investing in a Qatari company. And some are a mix of both, such as a UAE company and a British company acquiring another UAE company. Therefore, we separately identify purely domestic, purely cross-border and domestic-plus-cross-border deals. Thereafter, we calculate the number of participants on both sides of each deal, i.e., the acquirers and targets, and equally divide the deal value among the acquirers and targets. The classification of the participants into domestic, cross-border, or mixed ownership helps us to identify marginal effects at the dis-aggregated level in the analysis. Furthermore, we identify SWFs' participants using the unique identifier (in the BvD database, *BvD ID*) of each SWF, as provided by the BvD Zephyr database or by their name in SWFI's database. Additionally, we classify the industry type of the entities using the European Community Nomenclature of Economic Activities (NACE) Rev.2 industry classification system at the 1-digit level. This categorization includes Agriculture, Manufacturing, Construction, Trade, Transportation and Communication, Financial, Restaurants and Accommodation, Real Estate and Business

⁴The current analysis utilizes data for the first half of 2023. The next draft will incorporate data for the entire year.

⁵The initial download contained 4,379 deals, but some deals were excluded due to duplication in the Zephyr database or missing information.

services, Hydrocarbon industries, and Others.

We standardize the deal values by converting all currency denominations to USD for the analysis. Additionally, the data on industry specific nominal and real GDP is obtained from the IMF World Economic Outlook (IMF WEO) Database. We also use the global growth and oil price data from the IMF WEO as controls. To group the countries into different regions we use the IMF WEO classification. To identify investments in information and communication (ICT) industries, medium-high and high-technology manufacturing, and knowledge intensive services (KIS) we use the OECD definition and classification based on the statistical classification of economic activities in the NACE Rev.2 classification codes.⁶ We identify investments in green and renewable sectors using a combination of SIC codes and a textual analysis of deal descriptions for keywords associated with green industries and technology (such as solar, wind, geothermal, biofuels, energy efficiency, etc).

After undertaking the data cleaning steps mentioned above, we have identified over 13,700 unique deals with an approximate value of USD 2.5 trillion cumulative over the 2000-2023 period.⁷ Furthermore, around 70 percent of the total number of deals stem from cross-border investments (including both inward and outward investments). Additionally, deals involving SWFs account for about 40 percent of the total number of deals, with the values of these investments amounting to around 60 percent of the total investment value (encompassing both cross-border and domestic investments). The data indicates that the majority of investments are conducted by the UAE and Saudi Arabia (both in terms of number of deals and their value). Further details are provided in the Appendix I.⁸

⁶NACE Rev.2 and Standard Industrial Classification (SIC) codes are available in the databases.

⁷The data coverage is generally poor prior to 2004. The 2023 Zephyr data only covers January to June of 2023.

⁸Furthermore, in order to estimate the effect of cross-border and SWFs' investment on the economic complexity of GCC countries we use data on Economic Complexity Index (ECI), as compiled by Harvard Growth Lab. The Economic Complexity Index ranks countries based on the diversity and sophistication of their export portfolios, with high complexity indicating a wide range of specialized capabilities that enable the production of complex products. Note that the lower the index, the more complex the economy is. Therefore, any technological change driven by investments should have a negative effect on the ECI of the country. Results will be added in the next draft of the paper.

IV Empirical Methodology

To empirically investigate the relationship between real GDP growth and investment, we use the local projection method developed by Jord $\tilde{\text{A}}$ (2005). Specifically, the equation is:

$$(1) \quad y_{is,t+k} - y_{is,t-1} = \beta_k R_{is,t} + \theta X_{is,t} + \alpha_{is} + \gamma_t + \epsilon_{is,t}$$

where $y_{is,t}$ is the log of real GDP in country i , industry s , and year t ; $R_{is,t}$ is the investment (inward, outward, or domestic) for country i , industry s , and year t , divided by corresponding nominal GDP or country i , industry s , and year t . $X_{is,t}$ is a set of control variables, including lags of the dependent variable and the investment, percentage change in oil prices, and global growth; α_{is} denotes country-industry fixed effects, included to control for unobserved cross-country-industry heterogeneity. β_k , therefore, measures the effect of investment on real GDP over the horizon of $k = 0, 1, 2, \dots, K$.

Subsequently, we expand the analysis to investigate the role of SWF investment in fostering GCC economic growth and diversification by focusing on SWF investment only in the right-hand side of equation (1).

V Results

Leveraging our novel deal-level dataset of GCC investment, we present two sets of results. The first subsection explores stylized facts for GCC aggregate and industry-specific investment from 2000 to 2022. The second section then presents empirical evidence of GCC investment impact on real GDP growth derived from the Local Projections method detailed in Section IV.

V.A Stylized Facts

Since 2000, GCC countries have experienced relatively modest levels of inward cross-border investment, averaging around 2 percent of total GDP (Figure I). A shift in dynamics occurred after the global financial crisis (GFC), when investment declined to below 1 percent. The oil price plunge of 2014-2016 further exacerbated the decline in investment inflows, leading to a period of stagnation. Since 2018, inward cross-border investment to the GCC have increased, driven by both global push factors (seeking higher returns, flight to safety, supply chains diversification, growing focus on sustainability) and GCC-idiosyncratic pull factors (Figure II).

GCC countries have exhibited a relatively quick rebound from the pandemic, although the pace and strength of the recovery vary slightly across the region. Direct spillovers on the GCC economies from global and regional geopolitical developments have been muted, with the exception of increased oil price volatility. Mega-events such as Expo 2020 in the UAE and the World Cup in Qatar further bolstered the region's recovery and investment activity.

Additionally, since 2018, the GCC countries have begun amending their laws, simplifying regulations, and establishing dedicated agencies to further attract foreign capital. Saudi Arabia and Kuwait introduced new foreign investment laws in 2018, allowing 100 % ownership in some sectors (SA) and offering tax incentives (Kuwait), streamlined procedures, and simplified regulations. The UAE increased foreign ownership limits in specific sectors like retail, agriculture, and healthcare in 2018. In 2021, the UAE further liberalized by allowing 100 % ownership in most onshore economic sectors. Additionally, the UAE established a dedicated agency for attracting foreign direct investment and implemented "golden visas" offering long-term residency for investors and entrepreneurs. Bahrain and Qatar also amended their commercial companies law and investment law (respectively) in 2018. Qatar additionally established free zones like the Qatar Free Zones Authority with specific benefits for foreign investors. Bahrain focused on developing the Bahrain Investment Wharf (BIW) as a hub for foreign investors. Oman revised its foreign investment law in 2019, aiming to simplify procedures, strengthen investor protection, and offer greater clarity on regulations. They also continue to develop and promote their special economic zones (SEZs). All GCC countries have identified specific sectors (such as tourism, manu-

facturing, logistics, and knowledge-based economic sectors) for targeted investment promotion, while also offering incentives and support for foreign investors interested in these areas. As a result, the UAE and Saudi Arabia have witnessed the fastest year-over-year (YoY) increase in M&A deals (from Zephyr) between 2021 and 2023, with volumes rising by almost 10 percent.

Similar dynamics are observed in the GCC's outward cross-border investment, driven largely by SWF investments abroad, though with significant heterogeneity across countries (Figures IV, V, and IX). While the UAE, Saudi Arabia, and Qatar have been increasing their cross-border investment abroad since 2018, following the 2014-2016 decline, Oman and Kuwait's investments haven't followed similar trends. This highlights the varying approaches of GCC countries to foreign investment strategies.

Traditionally, cross-border investment inflows in the GCC were skewed towards the hydrocarbon sector. However, the past two decades have seen increasing investment in manufacturing, construction, and finance sectors. Recent trends reveal a significant shift towards investments in service sectors, encompassing industries like transportation and logistics, information technology, and professional and business services (Figure III). Data for 2023 suggests that GCC countries continue to attract investment despite global headwinds and recent geoeconomic tensions.

Mirroring their inward investment focus on non-oil sectors, GCC countries' outward investment have recently targeted manufacturing and the vibrant marketable services sector, which could act as a vital conduit for technology transfer and innovation (Figure VI). GCC SWFs, such as the Abu Dhabi Investment Authority (ADIA) and Mubadala from the UAE, as well as the Public Investment Fund (PIF) from Saudi Arabia (KSA), led deal activity in the GCC region during the first half of 2023. The technology sector contributed the most to the total deal volume, with a value of USD 15 billion, followed by the chemical sector and the provider care sector.

Since the 2000s, the GCC has attracted inward investment from more diversified destinations beyond its own region and the Middle East and Central Asia. The Western Hemisphere has emerged as the main destination for both GCC outward and inward investment since the pandemic (Figures III and VI). As a result, investors from the Western Hemisphere and European countries have recently equalized their

shares in GCC inward and outward investment. Additionally, there has also been an increase in investments and commitments within the GCC and ME&CA regions, reflecting stronger regional ties and integration.

As shown in Figures VII-X, SWFs are emerging as central players in the GCC's economic narrative. SWFs account for over 80 percent of the GCC's total outward investment and around 40 percent of its domestic investment in our sample, highlighting their crucial role in both economic diversification and regional integration.

V.B Empirical Results

The empirical results, employing the Local Projections econometric technique developed by Jordà (2005) (see Section IV for model specification), underscores the positive relationship between cross-border inward investment and real GDP in GCC countries (at country-industry level). Figure XIII(A) shows that a one percentage point increase in inward investment-to-GDP ratio is associated with an estimated 1.2 percent increase in real GDP after five years. This finding reinforces the economic benefits of attracting foreign investment in non-hydrocarbon sectors, underlining its role in stimulating economic growth and productivity. The results align with the global consensus on the crucial role of foreign investment (in particular FDI) as a catalyst for economic development and growth. Additionally, as we only focus on non-hydrocarbon sectors in the empirical analysis, the positive growth effect also highlights the role of inward investment in promoting non-hydrocarbon growth drivers and thus economic diversification.

Similarly, Figure XIII(C) shows the positive link between GCC domestic investment and real GDP (at the country-industry level). Our analysis finds that a one percentage point increase in domestic investment-to-GDP ratio is associated with an estimated 0.4 percent increase in real GDP after five years. The lower elasticity could be linked to the different nature of GCC domestic investment (these could include projects with longer time horizons, such as building a nuclear power plant in the UAE or NEOM project in Saudi Arabia) or their potentially lower efficiency.

Our analysis does not reveal a significant relationship between GCC outward in-

vestment, including those by GCC SWFs, and GCC GDP growth (Figure XIII(B) and Figure XIV(A)). However, it suggests that domestic investment by GCC SWFs play an important role in stimulating GCC growth and development (Figure XIV(B)). The impact of these SWFs' investment is similar in magnitude to the impact of overall domestic investment on the GCC growth (at country-industry level).

To further investigate the impact of GCC investment on economic growth and diversification, we leverage the OECD's classifications for medium-high and high-technology manufacturing and knowledge-intensive services. We conduct a modified Local Projections (LP) analysis using high and low technology investment as right-hand side variables in equation (1) from Section IV. This analysis incorporates an interaction term between the total investment share of GDP and a dummy variable equal to 1 for investment directed towards the medium-high and high-technology manufacturing and knowledge-intensive services sectors.⁹ Utilizing this modification of equation (1) provides us with the ability to interpret the coefficients in terms of their marginal impact.

While the literature suggests larger multiplier effects for ICT investments on growth, our results show no statistically significant difference between the impact of high-versus low-technology GCC outward or inward investments (in percent of GDP) on non-hydrocarbon growth in the GCC region (at the country-industry level) within our sample. This could be attributed to two factors, based on a closer examination of the data: (1) the relatively recent increase in the investments in high-technology sub-sectors in the GCC, and (2) potentially insufficient breakdown of the non-hydrocarbon GDP, for example, breakdown of overall GDP of the transportation, information and communication sector into the GDP of those sub-sectors that are high-technology and low-technology.

However, we do find a differentiated impact for SWF-involved outward high-technology investments compared to all other outward investments (in percent of GDP). The coefficient suggest that a one percentage point increase in SWF-involved outward high-technology investment to GDP is associated with an estimated 0.4 percent increase in GCC GDP growth within five years, with a three-year lag. While this effect is statistically significant, it is not robust to changes in model specifications or

⁹We also tried separate models for the manufacturing the services sectors.

different classifications of high- vs. low-technology investments. Notably, the positive impact seems to be driven by SWFs' investments in knowledge-intensive services sectors. The positive impact suggests potential benefits of technology transfer through outward investment. The lagged effect could be attributed to the complexity of knowledge transfer and its subsequent integration into the domestic GCC value chain. We also find some evidence that GCC inward investment is driven by outward investment from previous years.¹⁰

[We plan to conduct further empirical analysis to examine the effect of cross-border and domestic investment on GCC economic complexity, as measured by the Economic Complexity Index (ECI) developed by the Harvard Atlas of Economic Complexity. This analysis will explore how these investments contribute to the diversification and sophistication of the GCC economies. The results will be incorporated in the next version of the draft.]

VI Conclusion

Recent years have witnessed a surge in investment activity across the GCC, encompassing both domestic and cross-border investments, as GCC countries implemented wide-ranging reforms to achieve economic and social development (IMF, 2023a). This shift can be partly attributed to the GCC's ambitious development strategies, which prioritize the development of strategic sectors such as advanced manufacturing, digital technology, and the green economy, to achieve higher sustainable growth and economic diversification. We leverage an innovative deal-level investments database to investigate the impact of investments (inward, outward, and domestic) on growth and economic diversification in the GCC region. Our study contributes to the literature on the economic benefits of inward foreign investment for GCC economic transformation and elaborates on the role of GCC SWF investments in the region's growth model.

Since the pandemic, GCC countries have attracted a higher share of foreign invest-

¹⁰The correlation between inward and outward investments is positive and significant in some specifications.

ment (in percent of GDP) compared to both advanced economies (AEs) and emerging markets. This trend, driven by the UAE, Saudi Arabia, and to a lesser extent Kuwait (IMF, 2023a), has positioned the GCC more centrally within the global foreign direct investment network over the past decade. Our analysis reveals that GCC countries continue to attract capital from AEs, with the Western Hemisphere and Europe accounting for roughly 60 percent of their inward investment. At the same time, the share of GCC investment as a proportion of total investment in the GCC region remains stable at around 25 percent, underscoring the region's commitment to continued regional cooperation and integration. These same regions are targeted by the GCC for their outward investments. SWFs serve as the main channel for GCC outward investment, and their share has been steadily increasing. Recently, GCC countries have also expanded their outward investments to the ME&CA region more broadly, further promoting intra-regional collaboration. SWFs are emerging as a key driver of this development. Furthermore, SWFs play a crucial role in bolstering domestic investment, accounting for approximately 40 percent of the investment activity within our sample.

Another notable trend since the pandemic is the increase of GCC inward and outwards investments to the services sector, driven by investments to the transportation and logistics, information and telecommunication technology sector (ICT), business and professional services, and other marketable services sectors. The share of inward investments into the services sector increased from around 30 percent on average during 2000-2019 to around 70 percent during 2020-2023 (excluding financial services). This picture is also mirrored by the GCC outward investment, with a significant proportion of SWFs' investment going to services. Within the GCC region, majority of SWFs investments also allocated to the manufacturing and marketable services sectors. Moreover, recently, in addition to their domestic investments in renewable and clean energy projects, the GCC countries are increasing the cross-border investments in clean and renewable energy greenfield projects, as shown by Figure XII, including through the investments by SWFs. In particular, there is a notable increase in the GCC outward renewable greenfield investments in 2022-2023, driven in large by the UAE and Saudi Arabia SWFs. This underscores SWFs' important role in driving GCC economic transformation and regional cooperation to implement GCC growth and developmental strategies and ensure sustainable growth.

Our empirical analysis reaffirms the positive contribution of both inward and domestic investments to GCC GDP growth and economic diversification. Notably, inward investment demonstrates a three times greater impact on growth compared to domestic investments. While outward GCC investment does not significantly influence domestic GDP growth, the recent trend of SWFs concentrating on the services sector increases their exposure to potential slowdowns in global growth or domestic demand. However, the geographically diversified nature of SWF investments should provide some mitigation. Limited data on green and renewable energy investment deals within our sample prevented econometric analysis of their impact on GCC growth. Nevertheless, the recent surge in investments towards clean and renewable energy sources signifies a positive step towards a smooth energy transition for GCC countries. This trend aligns with recent IMF studies (2022, 2023b) that modeled the potential benefits of green investments for GCC growth. Similarly, we haven't identified statistically significant differences in the impact of increased investments into high-technology sub-sectors in the GCC. This could again be attributed to the relatively recent uptick in such investments. These findings suggest that GCC policymakers should continue pursuing policies that attract foreign investment into non-hydrocarbon sectors to foster sustainable growth, while maintaining domestic investments in these areas. Our study offers valuable insights for other resource-rich nations facing similar challenges of economic diversification and regional cooperation in the face of global issues like climate change and digital transformation.

[We are currently expanding the analysis to include all of 2023 data. Additionally, we are working on ensuring the robustness of our results for high- versus low-technology and value-added investments. The next version of the draft will also include an analysis of economic complexity, focusing on how cross-border and SWF-involved investments impact on GCC economic complexity. We look forward to presenting these additional findings in the next draft.]

References

Amighini, A., & Sanfilippo, M. (2014). Impact of South-South FDI and trade on the export upgrading of African economies. *World Development*, 64, 1-17.

Engel, D., & Kammen, D. M. (2009). Green jobs and the clean energy economy. *Copenhagen: Copenhagen Climate Council*.

Garrett-Peltier, H. (2017). Green versus brown: Comparing the employment impacts of energy efficiency, renewable energy, and fossil fuels using an input-output model. *Economic Modelling*, 61, 439-447.

Hepburn, C., O'Callaghan, B., Stern, N., Stiglitz, J., & Zenghelis, D. (2020). Will COVID-19 fiscal recovery packages accelerate or retard progress on climate change?. *Oxford review of economic policy*, 36(Supplement_1), S359-S381.

Hussein, M. A. (2009). Impacts of foreign direct investment on economic growth in the Gulf Cooperation Council (GCC) Countries. *International Review of Business Research Papers*, 5(3), 362-376.

International Monetary Fund (IMF), 2014. Economic Diversification in the GCC: Past, Present, and Future. *Staff Discussion Note*, Washington.

International Monetary Fund (IMF), 2022. Assessing the National Investment Strategy using the DIGNAR-19 model. *IMF Country* , No. 22/275.

International Monetary Fund (IMF), 2023a. Economic Prospects and Policy Priorities for the GCC Countries - 2023. *IMF Country Report*, No. 23/413.

International Monetary Fund (IMF), 2023b. UAE reforms for more sustainable and greener growth. *IMF Country Report*, No. 23/224.

Iwamoto, M., & Nabeshima, K. (2012). Can FDI promote export diversification and sophistication of host countries?: dynamic panel system GMM analysis (No. 347). *Institute of Developing Economies, Japan External Trade Organization (JETRO)*.

Jord  ,  . (2005). Estimation and inference of impulse responses by local projections. *American economic review*, 95(1), 161-182.

Lee, K. J., Fariza, B., & Sharipova, Z. (2015). The relationship between FDI, diversification and economic growth in natural resource-oriented countries: case of Kazakhstan. *Journal of International Business and Economics*, 3(2), 51-62.

Mami, E. (2023). The role of sovereign wealth funds in natural resource-rich countries: A systematic meta-narrative review. *Journal of International Business Policy*, 1-16.

Tadesse, B., & Shukralla, E. K. (2013). The impact of foreign direct investment on horizontal export diversification: empirical evidence. *Applied Economics*, 45(2), 141-

159.

OCED (2021). Understanding firm growth. Helping SMEs Scale up. *OECD Studies on SMEs and Entrepreneurship*, Annex C, 121-126.

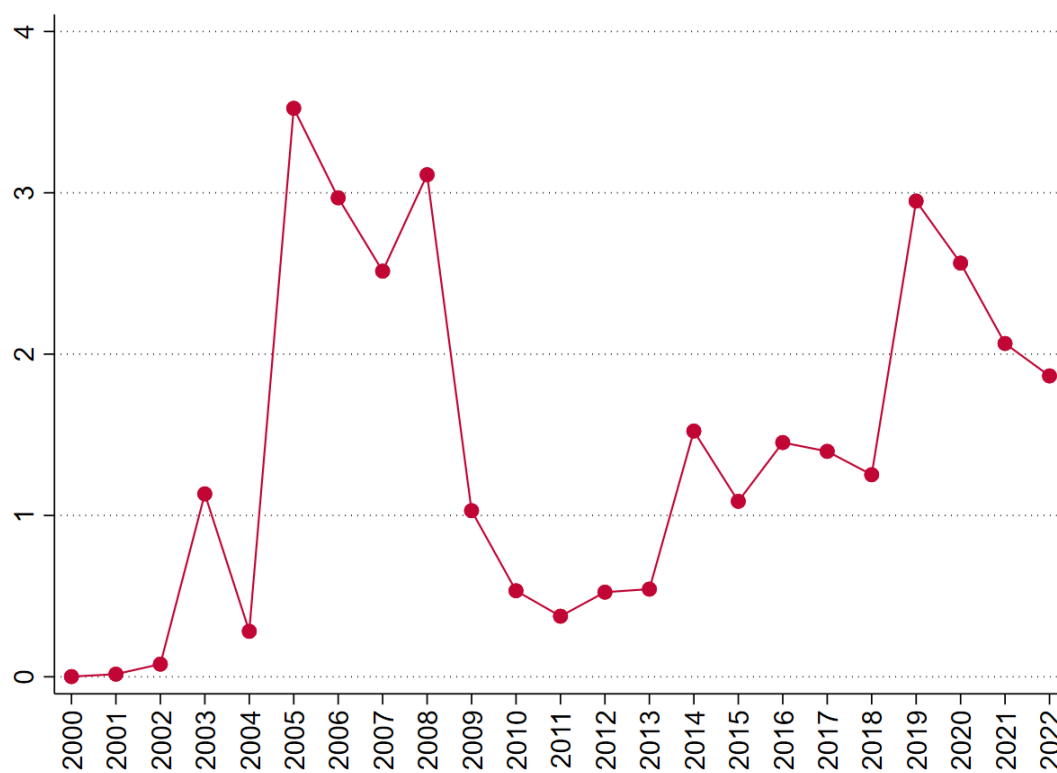
Appendix I: Tables and Figures

TABLE I: SUMMARY STATISTICS

Jurisdiction	Total Number of Deals	Total Deal Value	Cross Border Deals	Cross Border Deal Value	SWF Involved Deals	SWF Involved Deal Values	SWF Involved Cross Border Deals	SWF Involved Cross Border Deal Value
BHR	709	72	590	56.1	60	4.5	43	3.9
KSA	2310	686	1702	436	701	461.8	520	249.1
KWT	1855	178	1552	159	854	79.1	835	77.1
OMN	795	38.9	611	30.7	192	12.9	120	6.7
QAT	1046	338	924	293.3	616	231.5	566	205.4
UAE	6999	1190	6110	761.2	2651	668.6	2418	493.9
GCC	13714	2502.9	11489	1736.9	5074	1458	4502	1036.3

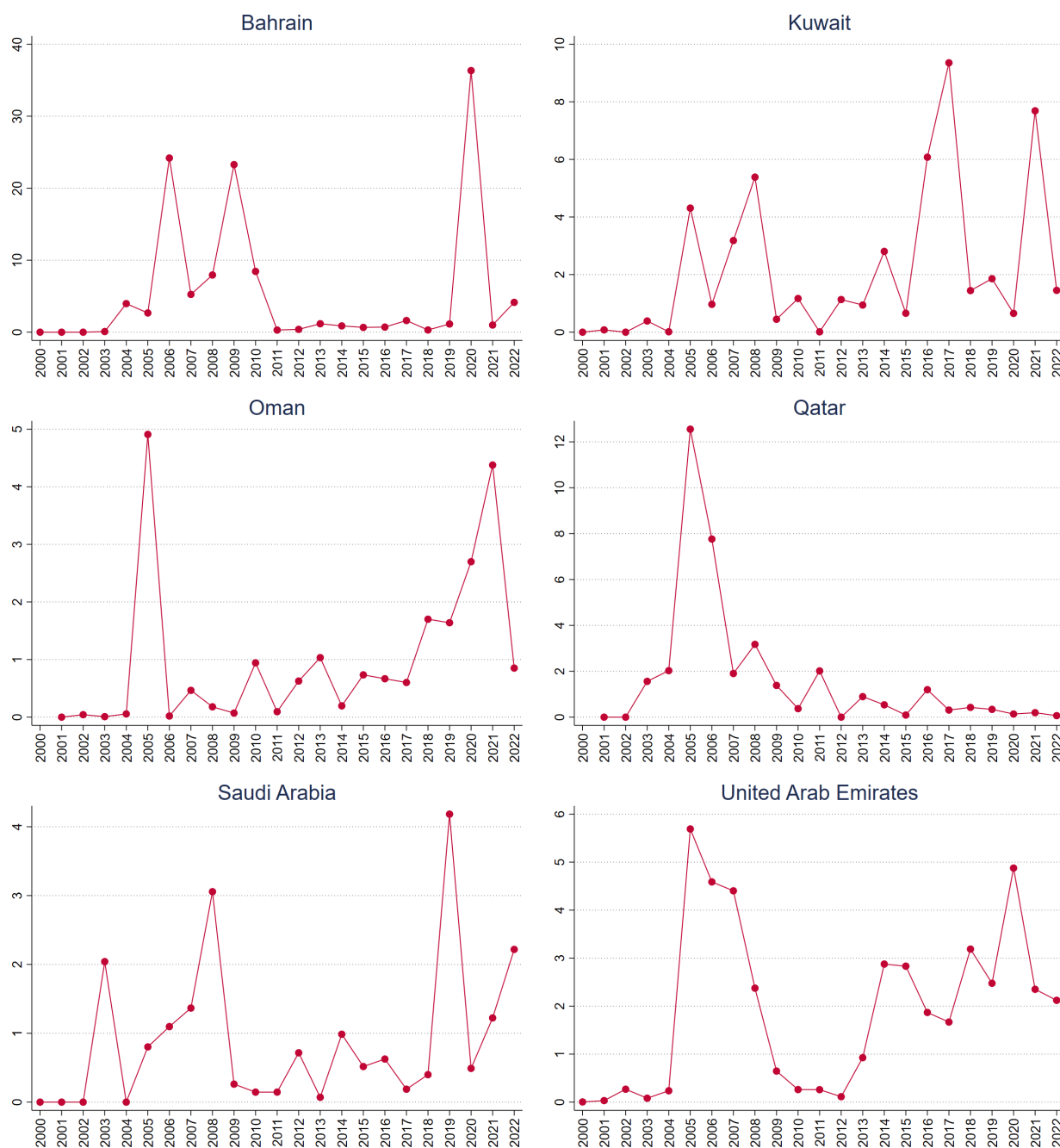
Notes: This table provides summary information for the database for the period 2000-2023. The first six rows provide country-specific deals' information, while the last row provides the cumulative information for the whole GCC region. All of the deal values are in nominal USD billion.

FIGURE I: GCC AGGREGATE: EVOLUTION OF INWARD INVESTMENT, IN PERCENT OF GDP



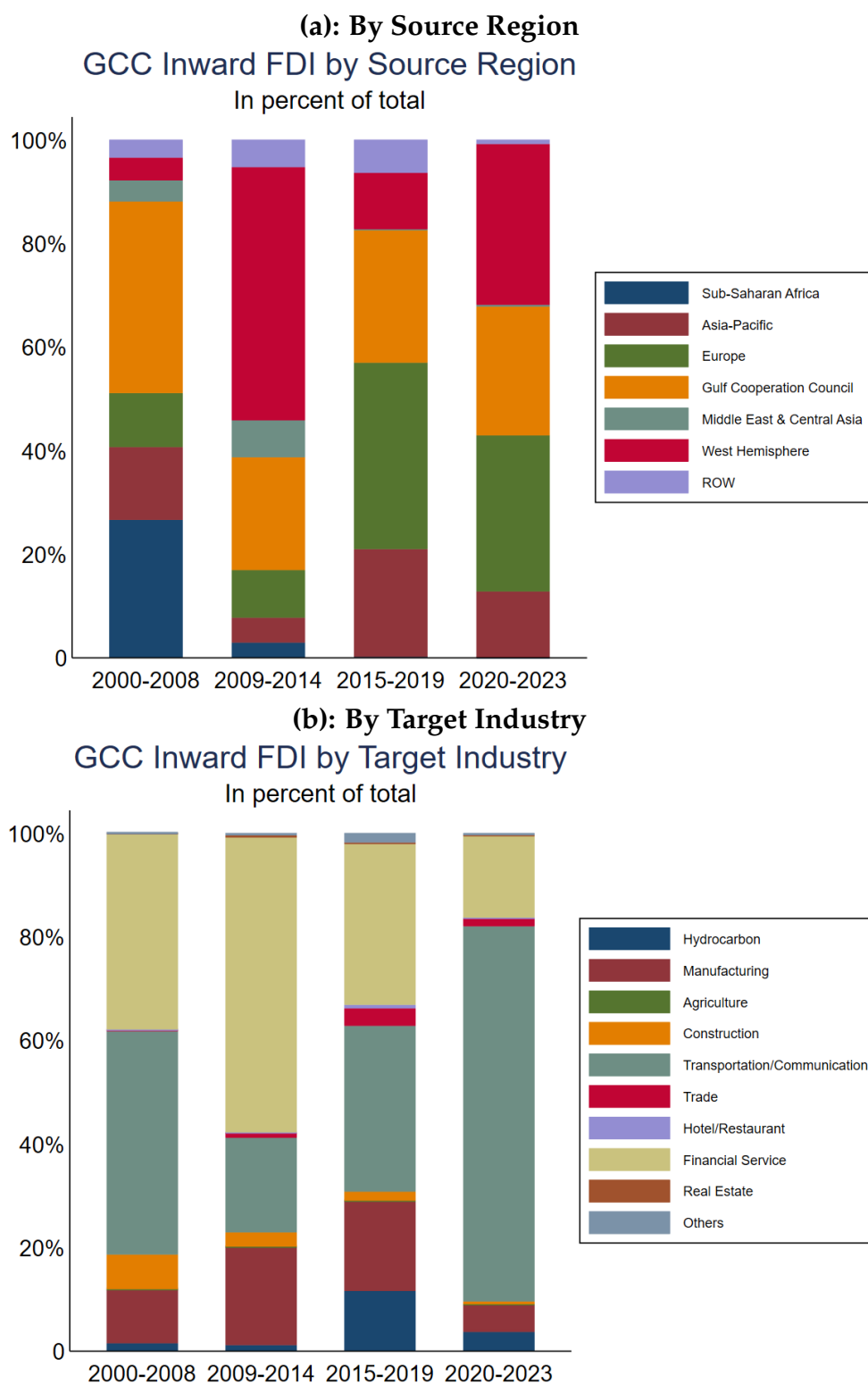
Notes: The figure plots the GCC total inward investment, in percent of GCC total GDP in each year during 2000-2022.

FIGURE II: GCC INDIVIDUAL COUNTRIES: EVOLUTION OF INWARD INVESTMENT, IN PERCENT OF GDP



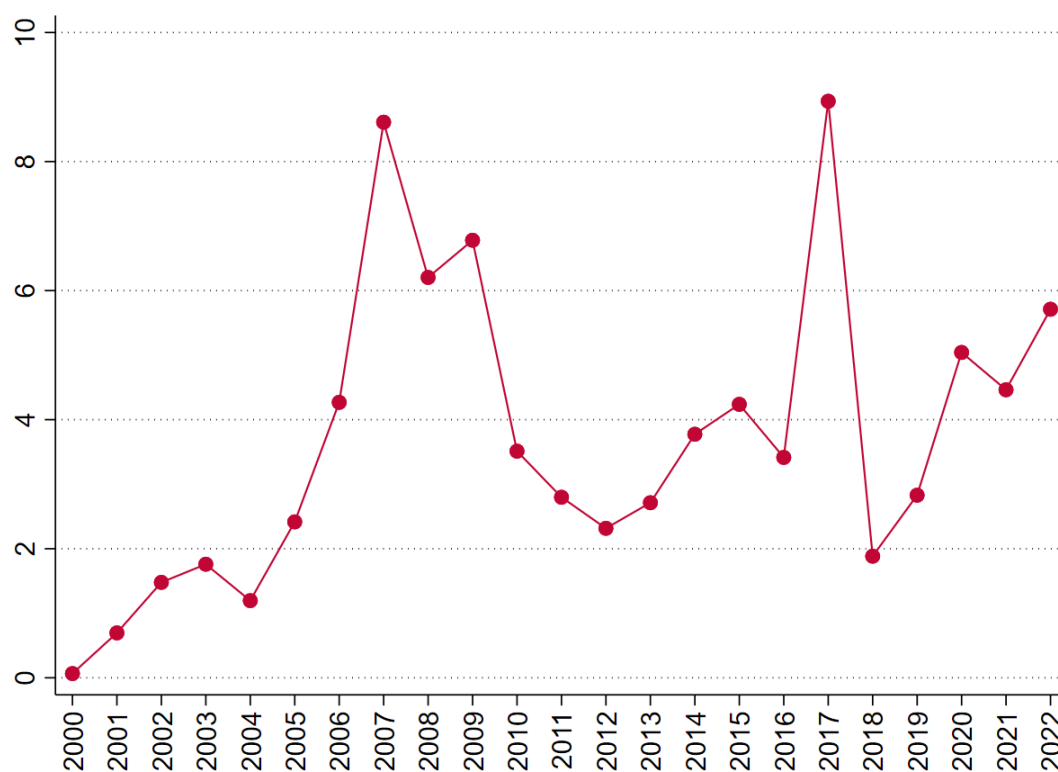
Notes: The figure plots each GCC country's inward investment, in percent of the country's GDP, in each year during 2000-2022.

FIGURE III: SOURCE REGION AND TARGET INDUSTRY OF GCC INWARD INVESTMENT



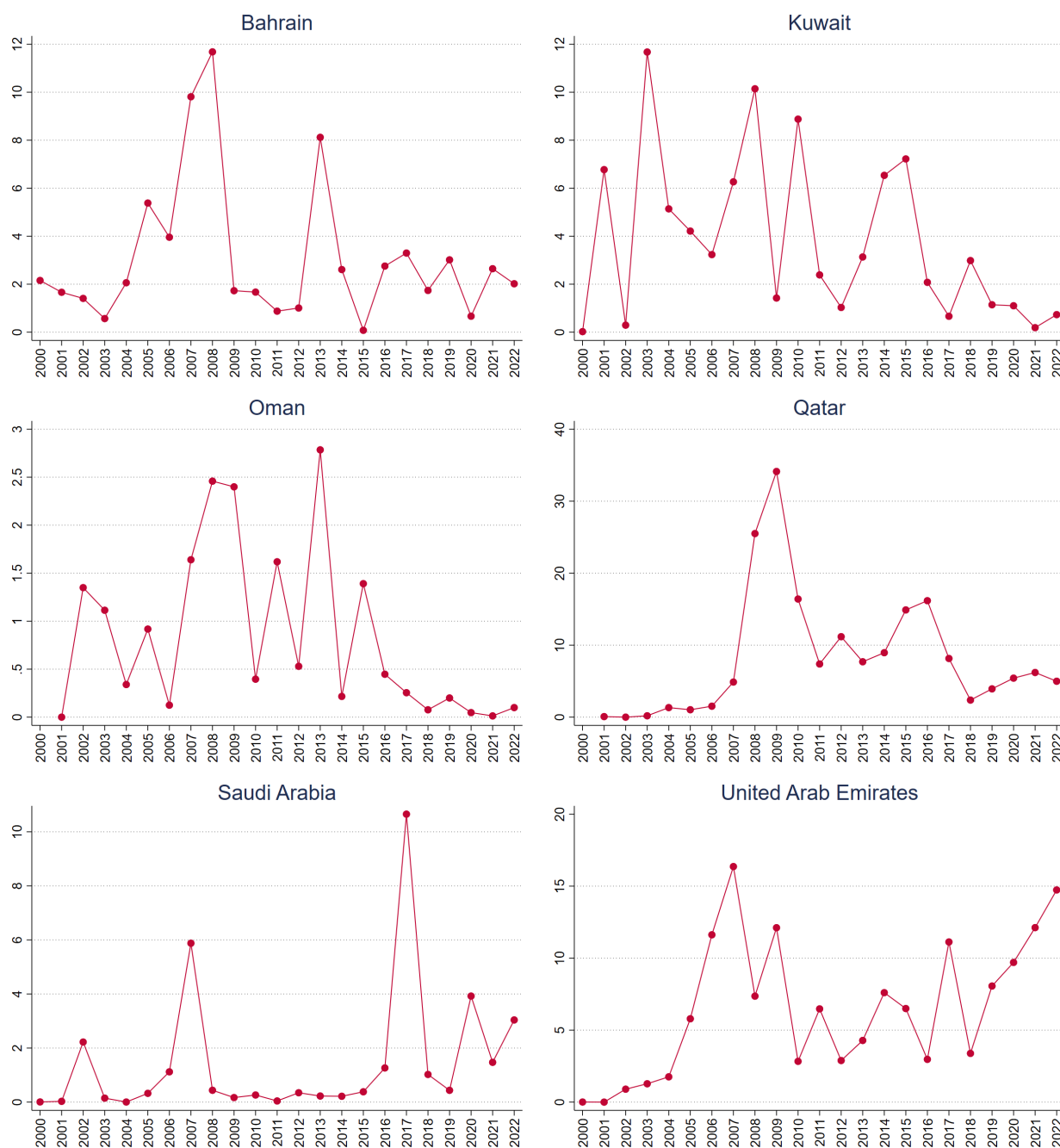
Notes: The figure (a) plots the share of inward investment from each source region, i.e., the region where the inward investment originates. The figure (b) plots the share of inward investment for each target industry, i.e., the industry in which the inward investment is received.

FIGURE IV: GCC AGGREGATE: EVOLUTION OF OUTWARD INVESTMENT, IN PERCENT OF GDP



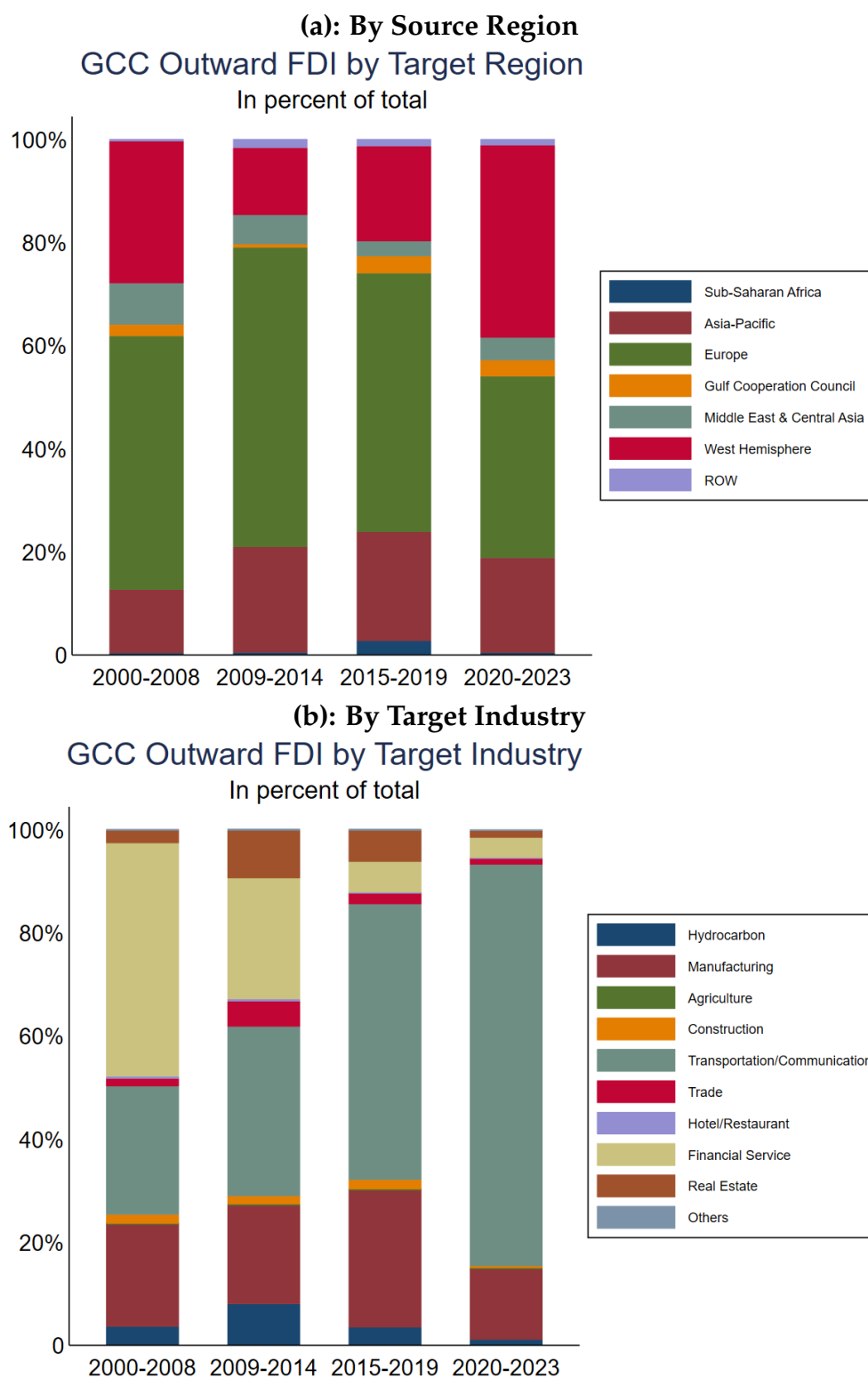
Notes: The figure plots the GCC total outward investment, in percent of GCC total GDP in each year during 2000-2022.

FIGURE V: GCC INDIVIDUAL COUNTRIES: EVOLUTION OF OUTWARD INVESTMENT, IN PERCENT OF GDP



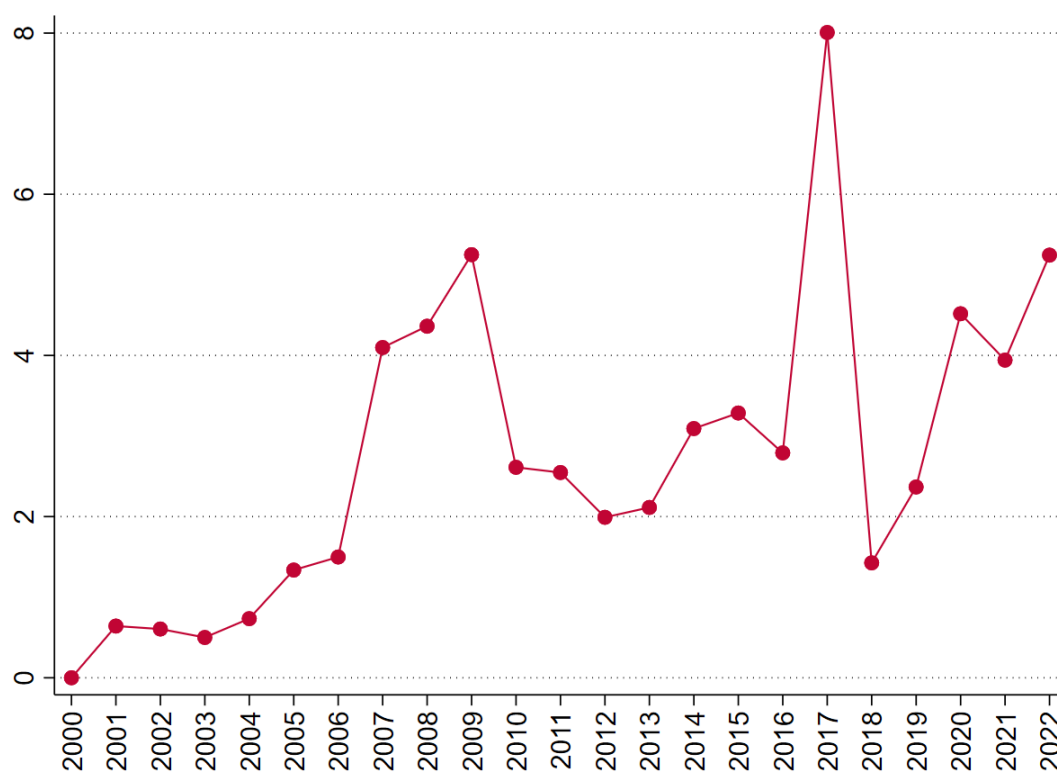
Notes: The figure plots each GCC country's outward investment, in percent of the country's GDP, in each year during 2000-2022.

FIGURE VI: TARGET REGION AND TARGET INDUSTRY OF GCC OUTWARD INVESTMENT



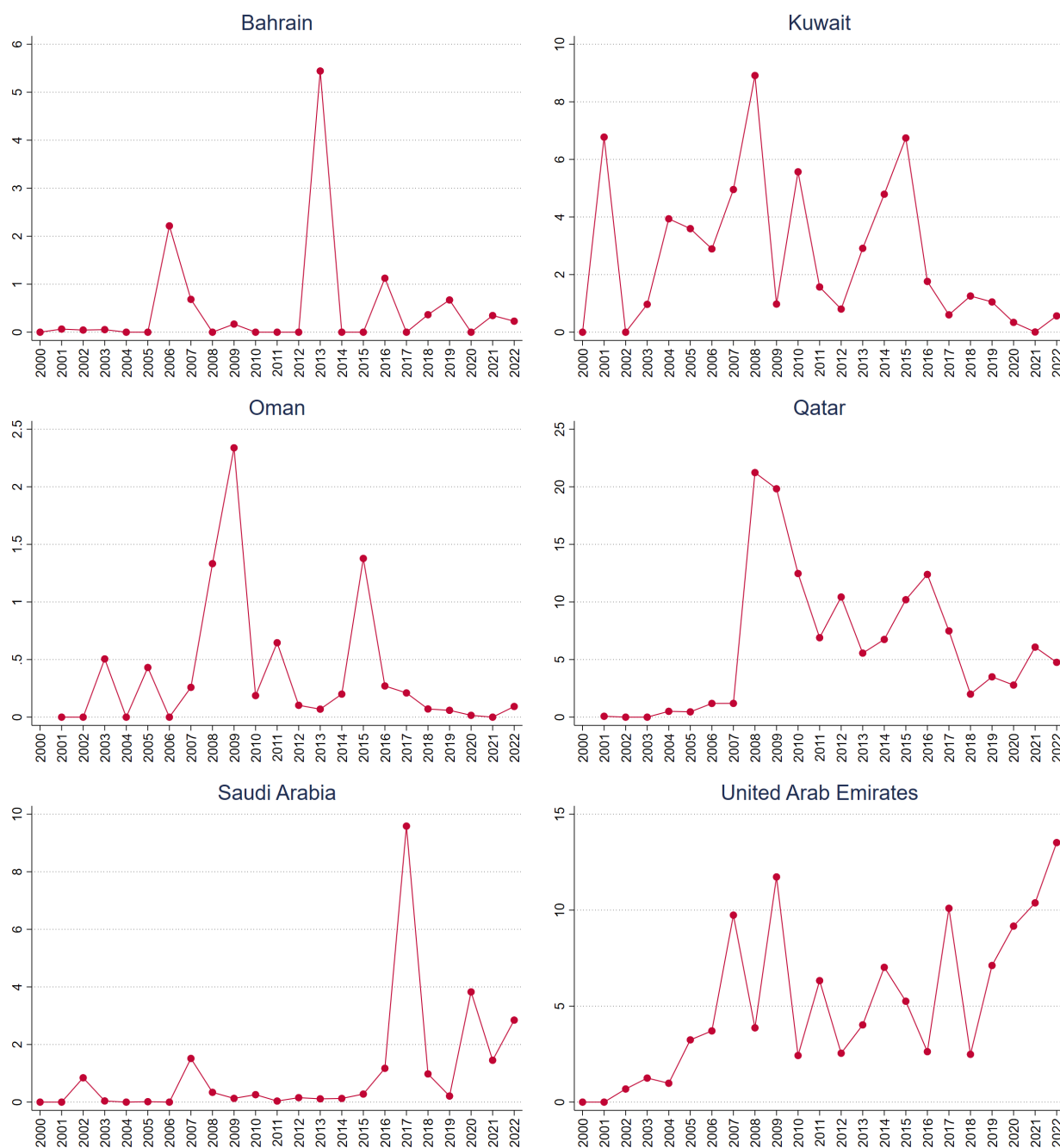
Notes: The figure (a) plots the share of outward investment from each target region, i.e., the region where the outward investment is sent. The figure (b) plots the share of outward investment for each target industry, i.e., the industry to which the outward investment is sent.

FIGURE VII: GCC AGGREGATE: EVOLUTION OF SWF OUTWARD INVESTMENT, IN PERCENT OF GDP



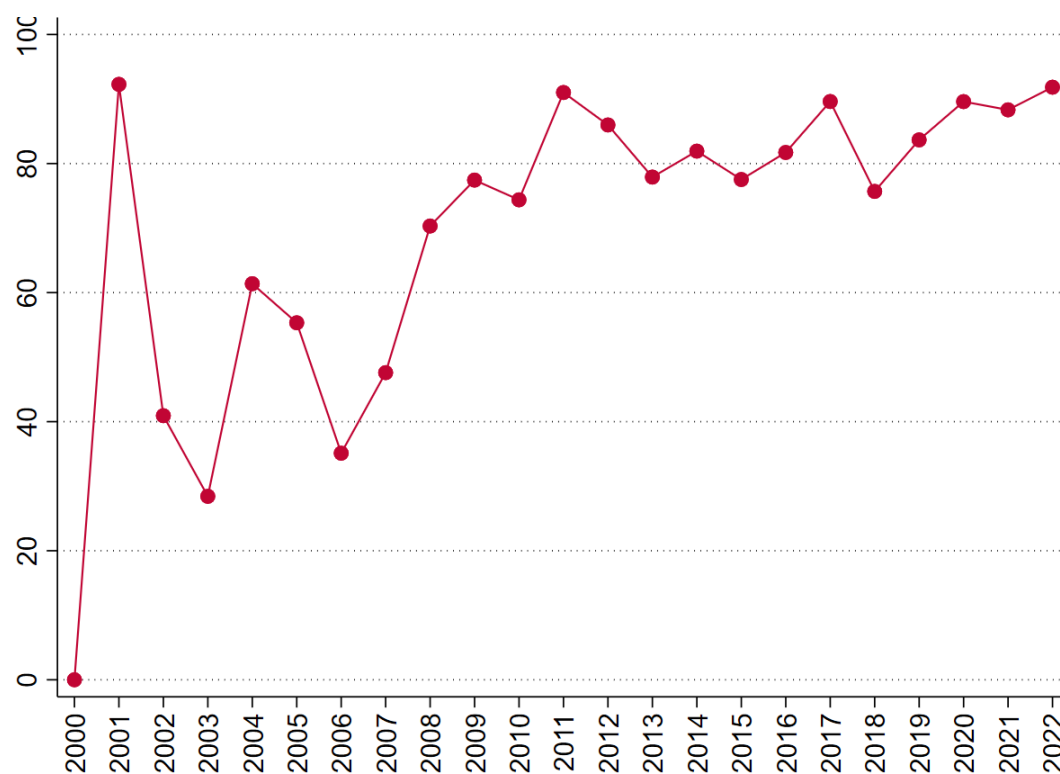
Notes: The figure plots the GCC SWF total outward investment, in percent of GCC total GDP in each year during 2000-2022.

FIGURE VIII: GCC INDIVIDUAL COUNTRIES: EVOLUTION OF SWF OUTWARD INVESTMENT, IN PERCENT OF GDP



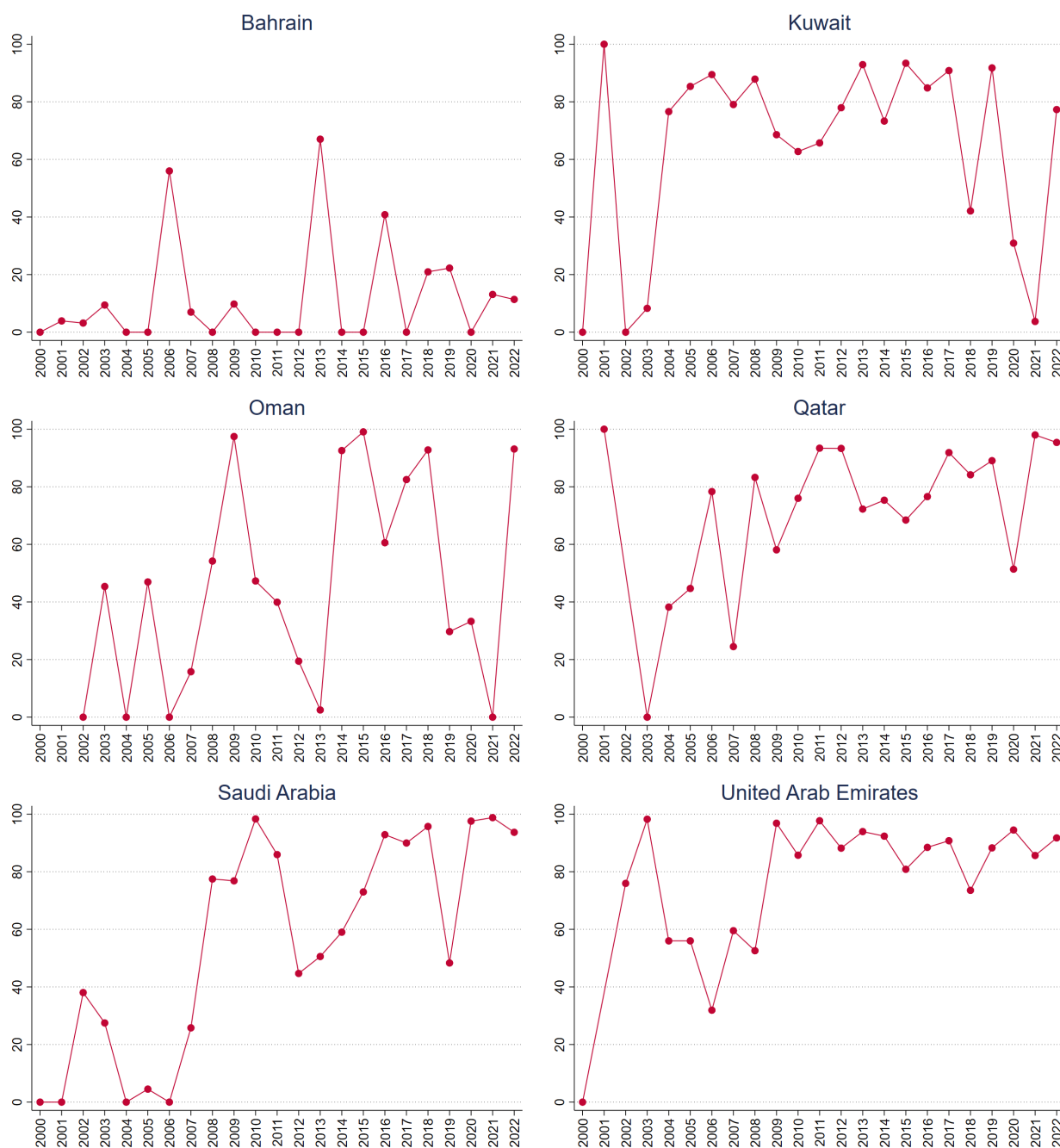
Notes: The figure plots each GCC country's SWF outward investment, in percent of the country's GDP, in each year during 2000-2022.

FIGURE IX: GCC AGGREGATE: EVOLUTION OF SWF OUTWARD INVESTMENT, IN PERCENT OF TOTAL OUTWARD INVESTMENT



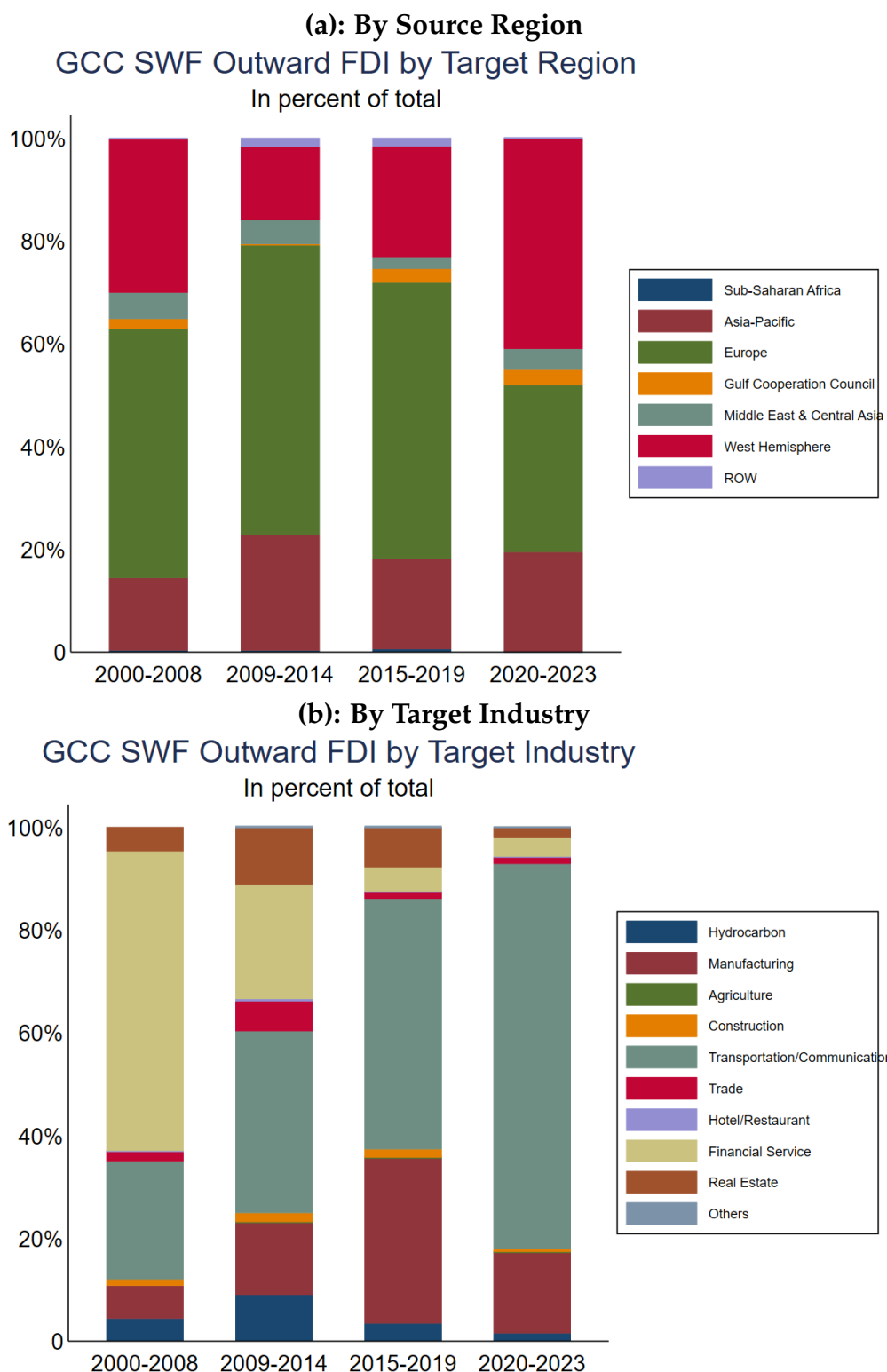
Notes: The figure plots the GCC SWF total outward investment, in percent of GCC total outward investment in each year during 2000-2022.

FIGURE X: GCC INDIVIDUAL COUNTRIES: EVOLUTION OF SWF OUTWARD INVESTMENT, IN PERCENT OF TOTAL OUTWARD INVESTMENT



Notes: The figure plots each GCC country's SWF outward investment, in percent of the country's GDP, in each year during 2000-2022.

FIGURE XI: TARGET REGION AND TARGET INDUSTRY OF GCC SWF OUTWARD INVESTMENT



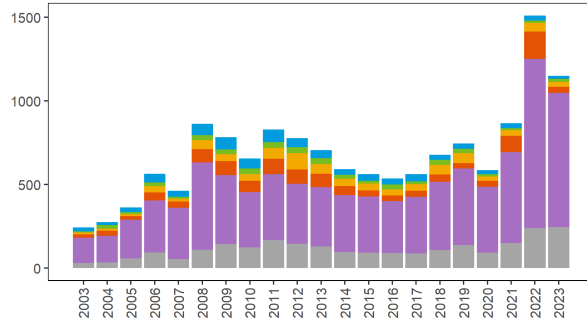
Notes: The figure (a) plots the share of SWF outward investment from each target region, i.e., the region where the outward investment is sent. The figure (b) plots the share of SWF outward investment for each target industry, i.e., the industry to which the outward investment is sent.

FIGURE XII: GCC GREENFIELD INVESTMENT

(A): Inward investment, number of deals

GCC Inward Greenfield Investment
(Number of Deals)

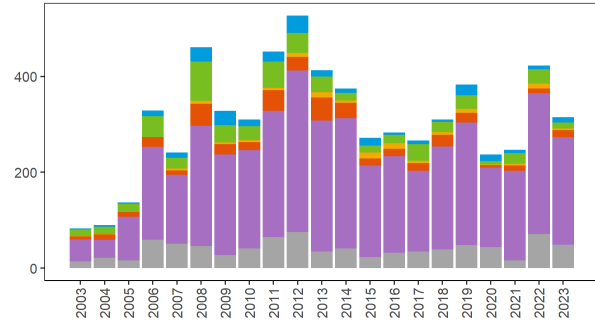
Bahrain Kuwait Oman Qatar UAE Saudi Arabia



(B): Outward investment, number of deals

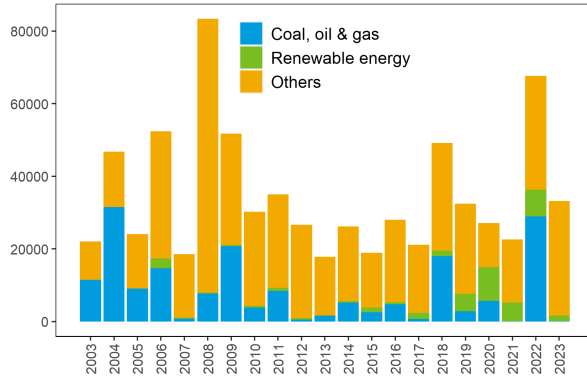
GCC Outward Greenfield Investment
(Number of Deals)

Bahrain Kuwait Oman Qatar UAE Saudi Arabia



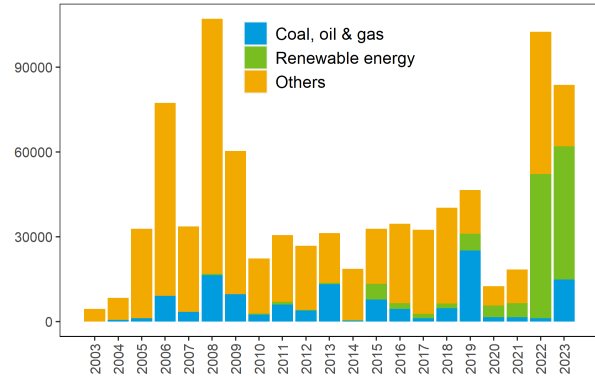
(C): Inward investment, sectors

GCC Inward Greenfield Investment
(Millions of US\$)



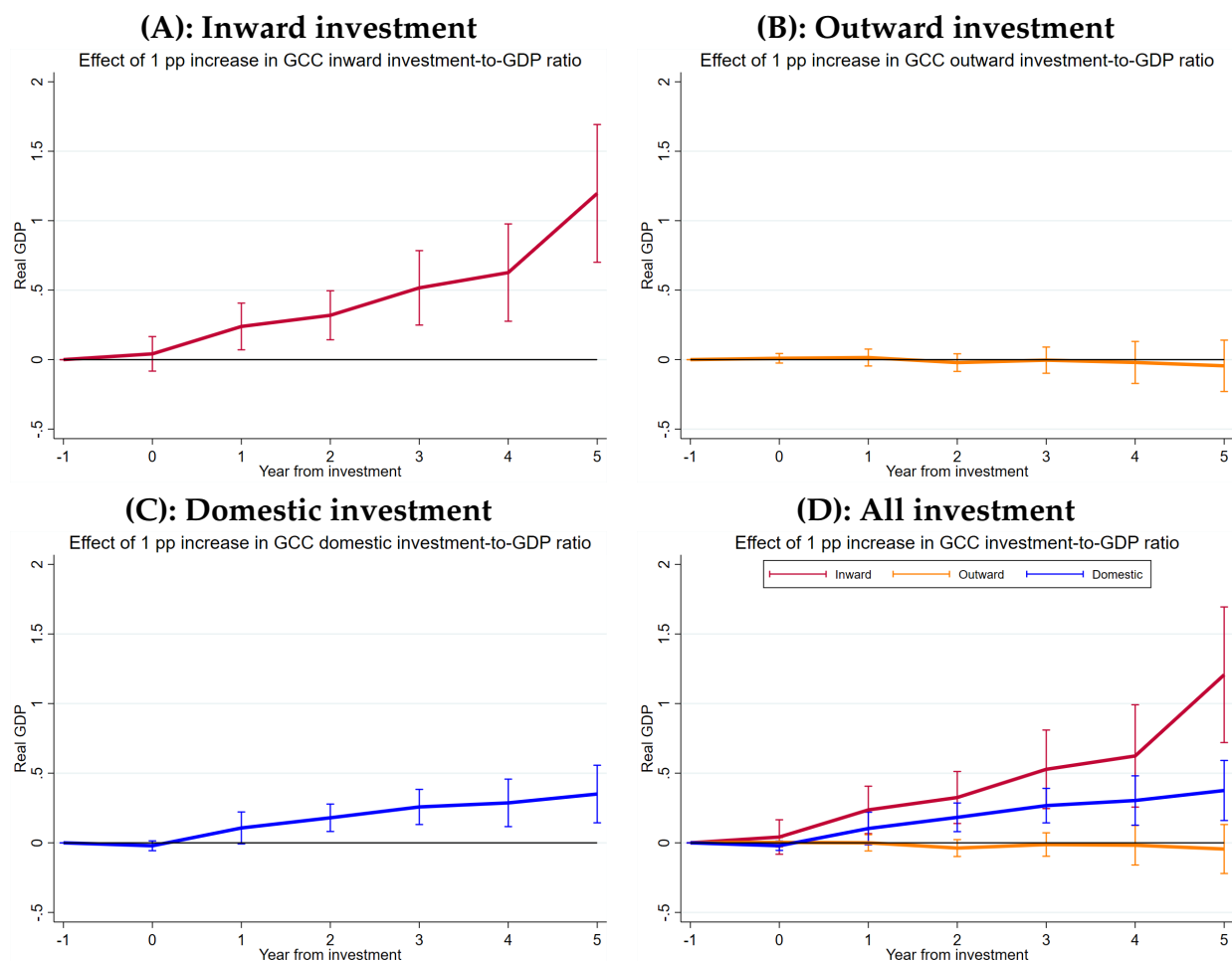
(D): Outward investment, sectors

GCC Outward Greenfield Investment
(Millions of US\$)



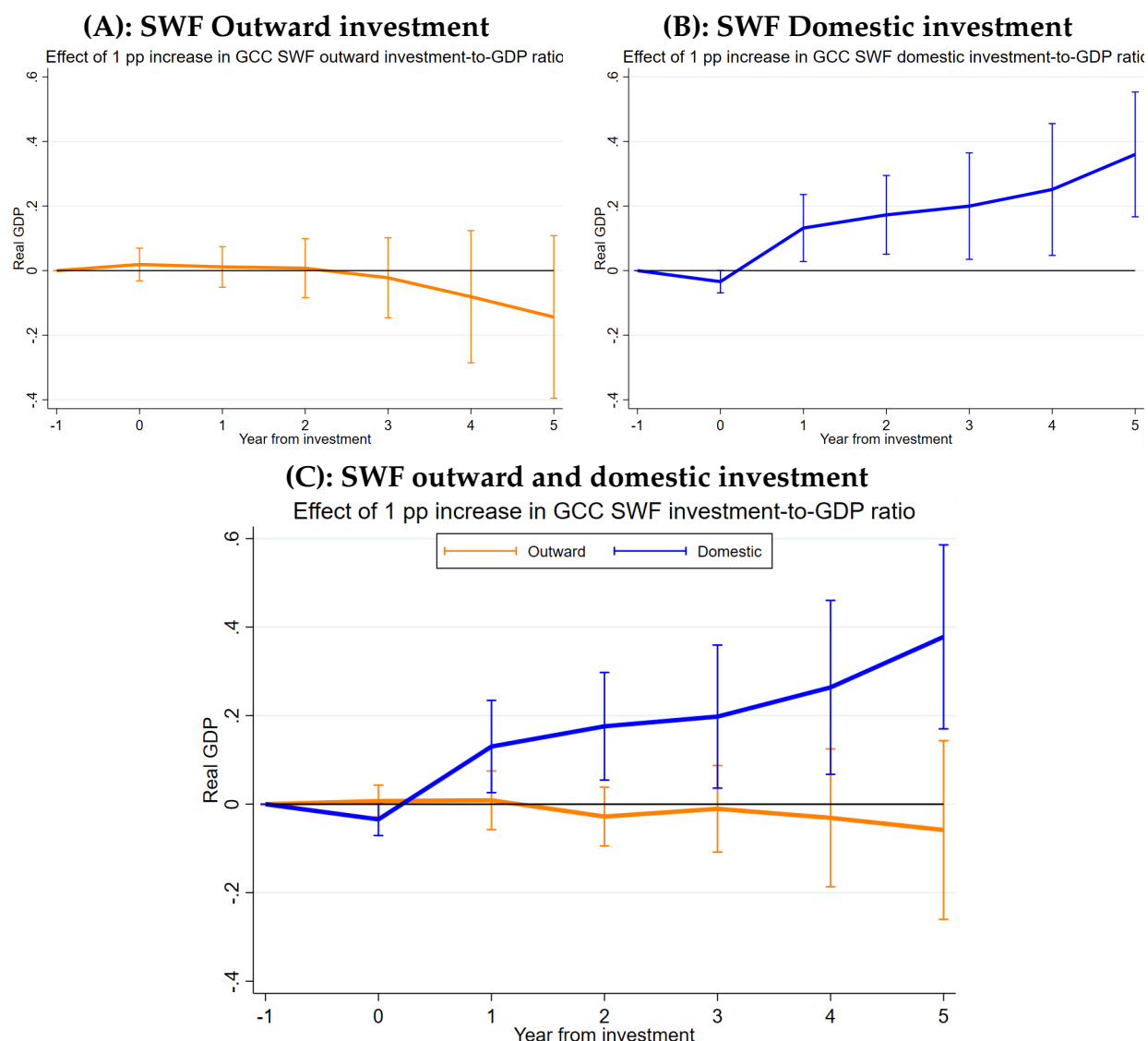
Notes: Figures (A) and (B) plot the number of inward and outward greenfield investment deals for GCC, by country. Figures (C) and (D) plot the deal values of inward and outward greenfield investment for GCC, separated by the target sector, namely, coal, oil, and gas, renewable energy, and others.

FIGURE XIII: LOCAL PROJECTIONS - GCC OVERALL INWARD, OUTWARD, AND DOMESTIC INVESTMENT



Notes: Figures (A)-(C) plot the coefficients estimated using Local Projections by Jorda (2005) as in equation (1) Section 3. Each figure shows the effect of 1 percentage point increase in the respective investment-to-GDP ratio on real GDP, in percent. Figure (D) plots the coefficients estimated using Local Projections by Jorda (2005) with all three kinds of investment included in the same equation, as robustness check. All the standard errors are clustered at the country-industry level.

FIGURE XIV: LOCAL PROJECTIONS - GCC SWF OUTWARD AND DOMESTIC INVESTMENT



Notes: Figures (A)-(C) plot the coefficients estimated using Local Projections by Jorda (2005) as in equation (1) Section 3. Each figure shows the effect of 1 percentage point increase in the respective SWF investment-to-GDP ratio on real GDP, in percent. Figure (D) plots the coefficients estimated using Local Projections by Jorda (2005) with both kinds of SWF investment included in the same equation, as robustness check. All the standard errors are clustered at the country-industry level.

TABLE II: LOCAL PROJECTIONS - GCC INWARD INVESTMENT

	(1) Year 0	(2) Year 1	(3) Year 2	(4) Year 3	(5) Year 4	(6) Year 5
Overall inward investment/GDP	0.042 (0.076)	0.239** (0.102)	0.319*** (0.107)	0.517*** (0.163)	0.626** (0.213)	1.196*** (0.301)
L.Overall inward investment/GDP	0.207** (0.081)	0.308*** (0.103)	0.486*** (0.146)	0.448** (0.192)	0.887*** (0.225)	1.063*** (0.243)
L2.Overall inward investment/GDP	0.067 (0.041)	0.218 (0.128)	0.242* (0.136)	0.513** (0.193)	0.670*** (0.191)	0.612** (0.239)
L.Real GDP growth	0.099 (0.061)	0.076 (0.160)	0.236 (0.141)	0.281 (0.200)	0.243 (0.246)	-0.024 (0.213)
L2.Real GDP growth	0.029 (0.101)	0.134 (0.126)	0.084 (0.121)	0.118 (0.173)	-0.053 (0.168)	0.021 (0.165)
Percentage change in oil prices	-0.000 (0.000)	-0.000 (0.000)	-0.001 (0.001)	-0.000 (0.001)	0.000 (0.001)	0.003*** (0.001)
Global real GDP growth	0.019*** (0.005)	0.017*** (0.004)	0.019*** (0.005)	0.025 (0.016)	0.017 (0.017)	-0.008 (0.015)
Constant	-0.020 (0.015)	0.016 (0.019)	0.045 (0.031)	0.062 (0.081)	0.141 (0.090)	0.272*** (0.077)
Within R2	0.180	0.094	0.084	0.061	0.069	0.166
Observations	578	536	494	452	410	368

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the results of estimating a parametric Local Projection models for GCC inward investment (in percent of GDP), as in equation (1) described in Section 3. The prefix L1 and L2 denote first and second lags, respectively.

TABLE III: LOCAL PROJECTIONS - GCC OUTWARD INVESTMENT

	(1) Year 0	(2) Year 1	(3) Year 2	(4) Year 3	(5) Year 4	(6) Year 5
Overall outward investment/GDP	0.010 (0.021)	0.015 (0.037)	-0.021 (0.039)	-0.004 (0.057)	-0.020 (0.092)	-0.045 (0.112)
L.Overall outward investment/GDP	0.025 (0.034)	-0.008 (0.033)	-0.002 (0.038)	-0.037 (0.058)	-0.046 (0.082)	0.027 (0.065)
L2.Overall outward investment/GDP	-0.005 (0.019)	0.007 (0.030)	-0.019 (0.052)	0.020 (0.053)	0.087 (0.064)	-0.009 (0.097)
L.Real GDP growth	0.099 (0.062)	0.080 (0.165)	0.241 (0.143)	0.316 (0.214)	0.298 (0.262)	0.018 (0.250)
L2.Real GDP growth	0.026 (0.102)	0.137 (0.129)	0.091 (0.125)	0.126 (0.174)	-0.053 (0.181)	0.043 (0.180)
Percentage change in oil prices	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)	0.003*** (0.001)
Global real GDP growth	0.019*** (0.005)	0.016*** (0.004)	0.018*** (0.005)	0.023 (0.015)	0.015 (0.017)	-0.011 (0.016)
Constant	-0.018 (0.014)	0.027 (0.016)	0.063** (0.028)	0.084 (0.077)	0.165* (0.090)	0.308*** (0.078)
Within R2	0.171	0.075	0.059	0.034	0.031	0.103
Observations	578	536	494	452	410	368

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the results of estimating a parametric Local Projection models for GCC outward investment (in percent of GDP), as in equation (1) described in Section 3. The prefix L1 and L2 denote first and second lags, respectively.

TABLE IV: LOCAL PROJECTIONS - GCC DOMESTIC INVESTMENT

	(1) Year 0	(2) Year 1	(3) Year 2	(4) Year 3	(5) Year 4	(6) Year 5
Overall domestic investment/GDP	-0.021 (0.021)	0.107 (0.070)	0.180*** (0.060)	0.258*** (0.077)	0.287** (0.104)	0.351** (0.126)
L.Overall domestic investment/GDP	0.119* (0.069)	0.210*** (0.072)	0.228 (0.157)	0.287 (0.175)	0.323* (0.163)	0.577*** (0.144)
L2.Overall domestic investment/GDP	0.086*** (0.025)	0.113* (0.064)	0.082 (0.110)	0.065 (0.098)	0.121 (0.113)	0.253** (0.106)
L.Real GDP growth	0.101 (0.062)	0.088 (0.159)	0.239 (0.150)	0.308 (0.216)	0.286 (0.273)	0.005 (0.247)
L2.Real GDP growth	0.031 (0.101)	0.146 (0.128)	0.097 (0.124)	0.136 (0.174)	-0.034 (0.179)	0.030 (0.171)
Percentage change in oil prices	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)	0.003*** (0.001)
Global real GDP growth	0.018*** (0.005)	0.015*** (0.004)	0.017*** (0.005)	0.022 (0.017)	0.014 (0.018)	-0.014 (0.017)
Constant	-0.017 (0.014)	0.024 (0.017)	0.057* (0.029)	0.079 (0.081)	0.162* (0.092)	0.305*** (0.079)
Within R2	0.176	0.086	0.066	0.040	0.034	0.119
Observations	578	536	494	452	410	368

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the results of estimating a parametric Local Projection models for GCC domestic investment (in percent of GDP), as in equation (1) described in Section 3. The prefix L1 and L2 denote first and second lags, respectively.

TABLE V: LOCAL PROJECTIONS - GCC INWARD, OUTWARD, AND DOMESTIC INVESTMENT

	(1) Year 0	(2) Year 1	(3) Year 2	(4) Year 3	(5) Year 4	(6) Year 5
Overall inward investment/GDP	0.042 (0.075)	0.236** (0.104)	0.325** (0.113)	0.528*** (0.171)	0.624** (0.223)	1.207*** (0.296)
L.Overall inward investment/GDP	0.198** (0.082)	0.305** (0.110)	0.497*** (0.151)	0.448** (0.197)	0.875*** (0.216)	1.066*** (0.244)
L2.Overall inward investment/GDP	0.065 (0.042)	0.215 (0.128)	0.238* (0.132)	0.499** (0.184)	0.649*** (0.194)	0.608** (0.246)
Overall outward investment/GDP	0.003 (0.020)	0.000 (0.035)	-0.038 (0.037)	-0.012 (0.051)	-0.017 (0.086)	-0.044 (0.107)
L.Overall outward investment/GDP	0.016 (0.031)	-0.020 (0.028)	-0.013 (0.034)	-0.039 (0.053)	-0.051 (0.079)	0.001 (0.065)
L2.Overall outward investment/GDP	-0.007 (0.018)	0.001 (0.028)	-0.025 (0.049)	0.012 (0.053)	0.078 (0.063)	-0.026 (0.079)
Overall domestic investment/GDP	-0.021 (0.021)	0.103 (0.071)	0.183*** (0.062)	0.267*** (0.075)	0.304** (0.108)	0.376** (0.131)
L.Overall domestic investment/GDP	0.111 (0.071)	0.204** (0.078)	0.234 (0.155)	0.304* (0.167)	0.362** (0.160)	0.582*** (0.147)
L2.Overall domestic investment/GDP	0.081*** (0.023)	0.114 (0.068)	0.097 (0.105)	0.077 (0.103)	0.137 (0.114)	0.259** (0.112)
L.Real GDP growth	0.099 (0.061)	0.084 (0.159)	0.233 (0.148)	0.280 (0.205)	0.237 (0.251)	-0.049 (0.214)
L2.Real GDP growth	0.034 (0.099)	0.142 (0.126)	0.100 (0.123)	0.124 (0.178)	-0.054 (0.174)	0.025 (0.162)
Percentage change in oil prices	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)	0.003*** (0.001)
Global real GDP growth	0.019*** (0.005)	0.016*** (0.004)	0.019*** (0.005)	0.024 (0.016)	0.016 (0.017)	-0.010 (0.015)
Constant	-0.022 (0.014)	0.013 (0.018)	0.043 (0.030)	0.062 (0.079)	0.137 (0.092)	0.272*** (0.081)
Within R2	0.187	0.104	0.094	0.070	0.082	0.185
Observations	578	536	494	452	410	368

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the results of estimating a parametric Local Projection models for GCC investment (in percent of GDP), as in equation (1) described in Section 3. The prefix L1 and L2 denote first and second lags, respectively.

TABLE VI: LOCAL PROJECTIONS - GCC SWF OUTWARD INVESTMENT

	(1) Year 0	(2) Year 1	(3) Year 2	(4) Year 3	(5) Year 4	(6) Year 5
SWF outward investment/GDP	0.019 (0.031)	0.011 (0.038)	0.008 (0.056)	-0.022 (0.075)	-0.081 (0.124)	-0.144 (0.153)
L.SWF outward investment/GDP	0.010 (0.028)	0.005 (0.048)	-0.032 (0.056)	-0.100 (0.094)	-0.116 (0.136)	-0.041 (0.155)
L2.SWF outward investment/GDP	0.001 (0.017)	-0.031 (0.032)	-0.076 (0.074)	-0.047 (0.086)	-0.025 (0.108)	-0.204 (0.123)
L.Real GDP growth	0.101 (0.064)	0.081 (0.165)	0.242 (0.142)	0.308 (0.209)	0.287 (0.260)	0.007 (0.260)
L2.Real GDP growth	0.027 (0.102)	0.139 (0.130)	0.088 (0.122)	0.124 (0.173)	-0.048 (0.181)	0.062 (0.170)
Percentage change in oil prices	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)	0.003*** (0.001)
Global real GDP growth	0.019*** (0.005)	0.016*** (0.004)	0.018*** (0.005)	0.022 (0.016)	0.014 (0.017)	-0.012 (0.016)
Constant	-0.017 (0.015)	0.028 (0.018)	0.065** (0.030)	0.091 (0.079)	0.178* (0.091)	0.322*** (0.082)
Within R2	0.170	0.076	0.062	0.037	0.032	0.111
Observations	578	536	494	452	410	368

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the results of estimating a parametric Local Projection models for GCC SWF outward investment (in percent of GDP), as in equation (1) described in Section 3. The prefix L1 and L2 denote first and second lags, respectively.

TABLE VII: LOCAL PROJECTIONS - GCC SWF DOMESTIC INVESTMENT

	(1) Year 0	(2) Year 1	(3) Year 2	(4) Year 3	(5) Year 4	(6) Year 5
SWF domestic investment/GDP	-0.034 (0.021)	0.132* (0.063)	0.173** (0.074)	0.200* (0.100)	0.251* (0.124)	0.360*** (0.117)
L.SWF domestic investment/GDP	0.120 (0.077)	0.200** (0.092)	0.169 (0.183)	0.248 (0.199)	0.322* (0.168)	0.590*** (0.138)
L2.SWF domestic investment/GDP	0.066* (0.037)	0.074 (0.075)	0.066 (0.121)	0.090 (0.109)	0.097 (0.133)	0.233* (0.132)
L.Real GDP growth	0.101 (0.062)	0.084 (0.160)	0.237 (0.151)	0.305 (0.217)	0.286 (0.275)	0.003 (0.249)
L2.Real GDP growth	0.028 (0.102)	0.146 (0.129)	0.096 (0.123)	0.134 (0.173)	-0.036 (0.176)	0.029 (0.170)
Percentage change in oil prices	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)	0.003*** (0.001)
Global real GDP growth	0.019*** (0.005)	0.015*** (0.004)	0.017*** (0.005)	0.022 (0.017)	0.013 (0.018)	-0.014 (0.017)
Constant	-0.016 (0.014)	0.026 (0.017)	0.059* (0.030)	0.081 (0.081)	0.165* (0.092)	0.309*** (0.079)
Within R2	0.174	0.083	0.063	0.038	0.033	0.119
Observations	578	536	494	452	410	368

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the results of estimating a parametric Local Projection models for GCC SWF domestic investment (in percent of GDP), as in equation (1) described in Section 3. The prefix L1 and L2 denote first and second lags, respectively.

TABLE VIII: LOCAL PROJECTIONS - GCC SWF OUTWARD AND DOMESTIC INVESTMENT

	(1) Year 0	(2) Year 1	(3) Year 2	(4) Year 3	(5) Year 4	(6) Year 5
SWF domestic investment/GDP	0.007 (0.022)	0.009 (0.040)	-0.028 (0.040)	-0.010 (0.059)	-0.031 (0.095)	-0.058 (0.123)
L.SWF domestic investment/GDP	0.022 (0.035)	-0.011 (0.034)	-0.003 (0.039)	-0.042 (0.059)	-0.051 (0.086)	0.026 (0.070)
L2.SWF domestic investment/GDP	-0.005 (0.020)	0.009 (0.031)	-0.018 (0.055)	0.022 (0.057)	0.100 (0.066)	0.018 (0.094)
scaled_dom	-0.034 (0.022)	0.130* (0.063)	0.176** (0.074)	0.198* (0.098)	0.264** (0.120)	0.378*** (0.126)
L.scaled_dom	0.116 (0.077)	0.201* (0.097)	0.172 (0.188)	0.262 (0.199)	0.366** (0.169)	0.602*** (0.147)
L2.scaled_dom	0.060 (0.040)	0.074 (0.086)	0.079 (0.126)	0.105 (0.122)	0.123 (0.154)	0.248 (0.159)
L.Real GDP growth	0.100 (0.062)	0.084 (0.163)	0.236 (0.151)	0.308 (0.219)	0.286 (0.272)	-0.007 (0.253)
L2.Real GDP growth	0.028 (0.101)	0.144 (0.129)	0.101 (0.125)	0.131 (0.175)	-0.047 (0.180)	0.035 (0.173)
Percentage change in oil prices	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)	0.003*** (0.001)
Global real GDP growth	0.019*** (0.005)	0.015*** (0.004)	0.017*** (0.005)	0.022 (0.016)	0.014 (0.018)	-0.014 (0.017)
Constant	-0.018 (0.014)	0.026 (0.017)	0.061** (0.028)	0.084 (0.078)	0.163* (0.092)	0.308*** (0.081)
Within R2	0.176	0.084	0.064	0.039	0.039	0.121
Observations	578	536	494	452	410	368

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the results of estimating a parametric Local Projection models for GCC SWF investment (in percent of GDP), as in equation (1) described in Section 3. The prefix L1 and L2 denote first and second lags, respectively.